Efficacy and Safety of Neurosyn in Relapsing multiple sclerosis: A Phase 3 Randomised, Double-Blind, Placebo-Controlled Trial

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Note: Entirely fictional study with invented drug and data, created for training. Do not use for clinical decisions.

Abstract

Background: Neurosyn (NSN) is a monthly subcutaneous anti-CD20 antibody under investigation for relapsing multiple sclerosis. **Methods:** In this multicentre trial, 904 adults were randomised 1:1 to Neurosyn or placebo for 52 weeks, stratified by baseline severity and region. The primary endpoint was annualised relapse rate. **Results:** The primary endpoint was 0.21 with Neurosyn versus 0.38 with placebo (p <0.001). Injection-site reaction occurred in 18.3% of Neurosyn patients and 6.0% of placebo patients. **Conclusions:** Neurosyn met its primary endpoint with a manageable safety profile in relapsing multiple sclerosis.

Methods

Eligible participants were aged 18 to 75 years with a confirmed diagnosis of relapsing multiple sclerosis and provided written informed consent. Exclusion criteria included pregnancy, severe hepatic impairment (Child-Pugh C), and concurrent use of strong CYP3A4 inducers. Randomisation used an interactive web-response system. The analysis followed the intention-to-treat principle with multiple imputation for missing data.

Results

Characteristic	Neurosyn	Placebo
Participants (n)	452	452
Mean age (years)	59.3	54.0
Female (%)	39	48
Primary endpoint	0.21	0.38
p-value	<0.001	-
Injection-site reaction (%)	18.3	6.0
Discontinuation due to AE (%)	4.7	1.1

Serious adverse events were reported in 11.5% of the Neurosyn group and 8.5% of the placebo group. There were 5 deaths in the Neurosyn arm and 4 in the placebo arm, none judged related to study treatment.

Discussion

The effect size observed with Neurosyn is clinically meaningful. Limitations include the 52-week duration, under-representation of patients aged over 75 years, and limited ethnic diversity. Long-term safety beyond one year requires an open-label extension, which is ongoing.

References

1. Synthetic reference list omitted. 2. Illustrative Guidelines on Management, 2025.